

PAFTP session, that PAFTP SDU SHALL be split into ordered, non-overlapping PAFTP segments so the set of all PAFTP segments may be reassembled into the original PAFTP SDU (see [Figure 29, “MATTERoPAF: Matter Message / PAFTP layering”](#)). Each PAFTP segment SHALL be prepended with a PAFTP packet header and sent in the Service Specific Info field inside the Service Descriptor Extension Attribute inside of a single [WFA-USD](#) SDF Follow-up message. If a PAFTP SDU is split into more than one PAFTP segment, the PAFTP segments SHALL be sent in order of their position in the original PAFTP SDU, starting with the PAFTP segment at the buffer’s head.

At any point in time, only one PAFTP SDU may be transmitted in each direction over a PAFTP session. The transmission of PAFTP segments of any two PAFTP SDUs SHALL NOT overlap. If the application attempts to send one PAFTP SDU while transmission of another PAFTP SDU is in progress, the new PAFTP SDU SHALL be appended to a first-in, first-out queue. The next PAFTP SDU SHALL be dequeued from this queue and transmitted once transmission of the current PAFTP SDU completes, that is, once all PAFTP segments of the current PAFTP SDU have been transmitted and received by the peer.

The first PAFTP segment of a PAFTP SDU sent over a PAFTP session SHALL have its Beginning Segment header flag set to indicate the beginning of a new PAFTP SDU (see [Table 37, “PAFTP Packet PDU format”](#)). The presence of this flag SHALL indicate the further presence of a 16-bit unsigned integer field, the Message Length, that provides the receiver with the total length of the PAFTP SDU. The last PAFTP segment for a given PAFTP SDU SHALL have its Ending Segment flag set to indicate the end of the transmitted PAFTP SDU. A PAFTP packet that bears an unsegmented PAFTP SDU—that is, a PAFTP SDU small enough to fit into a single PAFTP segment—SHALL have both its Beginning Segment and Ending Segment flags set.

The size of a single PAFTP SDU sent via PAFTP is limited to 64KB, that is, the maximum size of the Message Length field in the PAFTP packet header. The number of segments used to send a buffer is unlimited and delimited by the Beginning Segment and Ending Segment bits in the PAFTP packet header. The upper layer imposes more stringent requirements over the maximum SDU size, such as [Section 4.4.4, “Message Size Requirements”](#).

The length of the data payload in each PAFTP segment whose Ending Segment bit is not set SHALL be equal to the session’s maximum PAFTP packet size minus the size of that PAFTP packet’s header. If a packet’s Ending Segment bit is set, the length of its PAFTP segment data payload SHALL equal the size of the original PAFTP SDU minus the total size of all previously transmitted PAFTP segments of that PAFTP SDU. In this way, the length of a SDU’s last PAFTP segment is implied by its size.

A peer SHALL reassemble PAFTP segments in the ascending order of the segment sequence numbers. Once a peer receives a complete set of PAFTP segments, it SHALL verify that the reassembled PAFTP SDU’s total length matches that specified by the Beginning Segment’s Message Length value. If they match, the receiver SHALL pass the reassembled PAFTP SDU up to the next-higher-layer. If the reassembled buffer’s length does not match that specified by the sender, or if received PAFTP segment payload size would exceed the maximum PAFTP packet size, or receiver receives an Ending Segment without the presence of a previous Beginning Segment, or a Beginning Segment when another PAFTP SDU’s transmission is already in progress, the receiver PAFTP SHALL close the PAFTP session and report an error to the application.